

KAPTURE FINANCE

Collections Voicebot — “Maya”

High-Level Design (HLD)

Outbound EMI-Collections Voice Agent — Architecture, Conversation Design & Compliance

AI Delivery Intern — Take-Home Assignment, Task 1
Prepared for: Kapture CX | Target build platform: Vapi
Version 1.0

0. Scenario & Design Intent

Kapture Finance needs an outbound voice agent (“Maya”) that calls customers with an overdue personal-loan EMI, verifies identity, discloses the overdue amount, negotiates a resolution, and hands off to a human only when the conversation genuinely requires judgement — all inside RBI-aligned fair-collection practice (no threats, no disclosure to third parties, permitted calling hours, honoring do-not-call requests).

Design principle that shapes every section below: authentication and disclosure are enforced by code, not by prompt instruction. The LLM proposes what to say; a deterministic state-machine layer decides what it is allowed to say. This is the single most important control in the system — it is what stops a model from being “talked past” verification under conversational pressure.

Reference case used throughout this document: Rahul Sharma, personal loan, overdue EMI ₹8,499, 12 days past due.

1. Architecture & Pipeline

The pipeline is a standard real-time voice stack (telephony → STT → orchestrator/LLM → TTS) with a tools layer for account, payment, and compliance systems, and an observability layer that taps every stage.

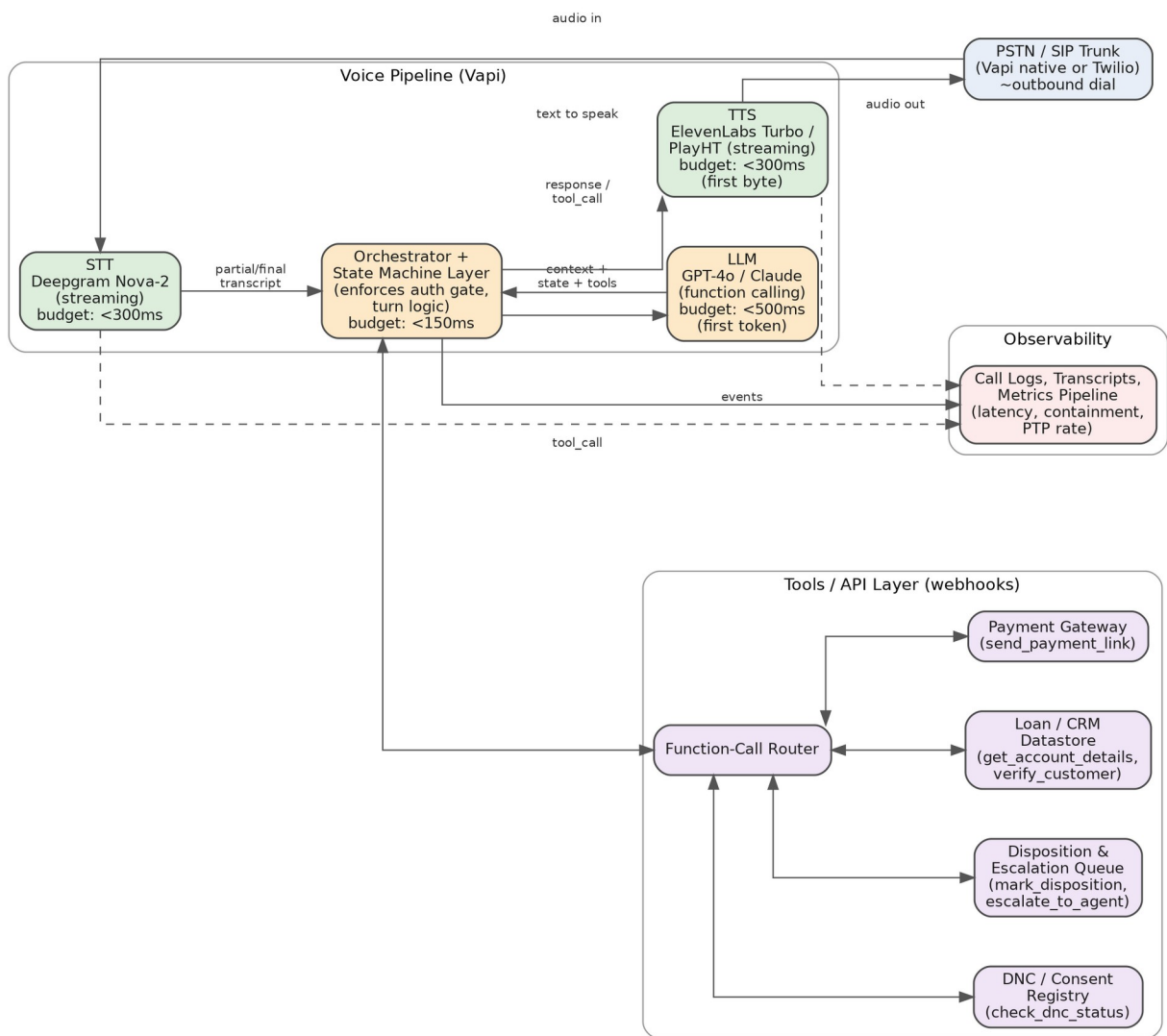


Figure 1. End-to-end pipeline: telephony, voice pipeline, tools/API layer, and observability.

1.1 Latency budget per hop

Target: under ~1.2s perceived turn latency (end of caller speech to start of bot audio), with barge-in enabled so the caller can interrupt at any time.

Hop	Component	Budget	Notes
1	STT (streaming)	< 300 ms	Deepgram Nova-2 (or Vapi built-in), partial transcripts streamed continuously; endpointing tuned to Hindi/English code-switch pauses.
2	Orchestrator / state check	< 150 ms	Deterministic layer: resolves current state, injects only the context/tools valid for that state before calling the LLM.
3	LLM first token	< 500 ms	GPT-4o / Claude with function calling; small, state-scoped system prompt (not the full flow) keeps prompt processing fast.
4	Tool call round-trip (when triggered)	< 400 ms	Mocked webhooks return synchronously; orchestrator plays a filler ("one moment") if a call exceeds ~600 ms.
5	TTS first byte (streaming)	< 300 ms	ElevenLabs Turbo / PlayHT streaming synthesis, sentence-chunked so audio starts before the full reply is generated.
6	Telephony egress	< 100 ms	Vapi-managed SIP/PSTN leg.

2. Conversation Flow / State Machine

The flow is modelled as an explicit finite-state machine that lives outside the LLM. Each state defines: what tools/context are exposed to the model, what the model is allowed to say, and which transitions are legal. Authentication (S2) is the hard gate: no state past S3 is reachable unless the orchestrator's own boolean `auth_verified` flag — set only by a successful `verify_customer()` tool result — is true. The LLM cannot set this flag itself; it can only trigger the tool call that may set it.

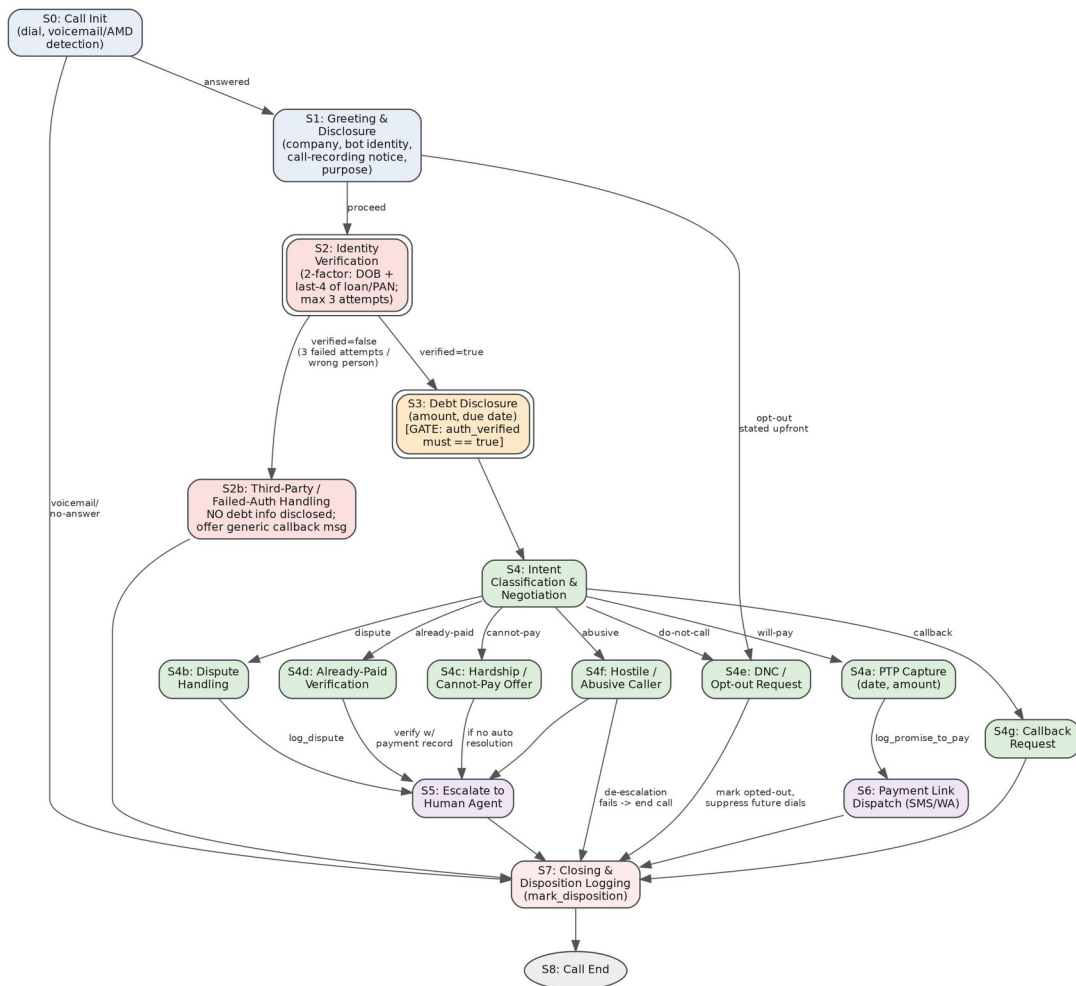


Figure 2. Full state machine. Double-bordered states are code-enforced gates.

2.1 State reference

State	What locks it / gate	Exit transitions
S0 Call Init	AMD (answering-machine detection) result	Answered → S1 · Voicemail/no-answer → S7 (leave compliant voicemail disclosure only, no amount)
S1 Greeting & Disclosure	Scripted, must complete before any question is asked	→ S2, or → S4e if caller opts out immediately
S2 Identity Verification	2-factor match (DOB + last-4 of loan a/c or PAN) scored server-side by <code>verify_customer</code> ; 3-attempt cap enforced by orchestrator counter, not by the model	verified=true → S3 · verified=false after 3 tries or explicit “wrong person” → S2b
S2b Third-Party / Failed Auth	Code-level block: response filter strips any account/amount tokens from LLM output in this state	→ S7 (disposition: <code>no_disclosure_unverified</code>)
S3 Debt Disclosure	GATE: only reachable with <code>auth_verified == true</code>	→ S4
S4 Intent Classification	Runs an intent-tagging pass on each user turn	Routes to S4a–S4g based on tagged intent
S4a–g sub-states	See §3 Intents & Entities	Each resolves to S5 (escalate), S6 (payment link), or directly to S7
S5 Escalate to	Triggered on dispute, unresolved hardship, repeated hostility, or	→ S7

State	What locks it / gate	Exit transitions
Human	model uncertainty above a confidence threshold	
S6 Payment Link Dispatch	Only after explicit verbal PTP with date + amount captured	→ S7
S7 Closing & Disposition	Every path must pass through here — no silent call termination	→ S8
S8 Call End	Terminal	—

3. Intents & Entities

3.1 Intents

Intent	Description	Example utterance	Routes to
will_pay	Customer agrees to pay and commits to a date/amount	“I’ll pay by Friday.”	S4a PTP Capture
cannot_pay / hardship	Customer states inability to pay — job loss, medical, income shock	“I lost my job, I can’t pay right now.”	S4c Hardship Offer
dispute	Customer disputes the amount, EMI count, or that a loan exists	“This isn’t my loan / the amount is wrong.”	S4b Dispute Handling
already_paid	Customer claims payment already made	“I paid this on the 3rd.”	S4d Already-Paid Verification
wrong_person	Person answering is not the customer	“There’s no Rahul here.”	S2b (pre-auth) or S7 (post-auth mis-route)
callback_request	Customer wants to be called at a different time	“Call me after 6pm.”	S4g Callback Request
hostile / abusive	Aggressive, abusive, or threatening language toward the bot	Raised voice, insults, refusal to engage	S4f Hostile Handling
do_not_call	Explicit opt-out / do-not-call request	“Stop calling this number.”	S4e DNC / Opt-out
language_switch	Mid-call switch between English and Hindi	Code-mixed or full switch	Handled in place — not a state change
out_of_scope	Unrelated question or off-topic request	“Can you help me apply for a new loan?”	Redirect once, then escalate if repeated

3.2 Entities extracted

Entity	Type	Extraction notes
auth_dob / auth_id_last4	string/date	Captured only inside S2; never logged in plaintext transcript — masked as ‘●●●●’ in logs, raw value passed directly to verify_customer over the tool channel.
ptp_date	date	Normalized to ISO-8601; relative expressions (“next Friday”, “mahine ke end tak”) resolved against call timestamp + IST.
ptp_amount	currency (INR)	Defaults to full overdue amount if unspecified; partial-payment amounts captured explicitly and flagged for review if < 50% of due.
dispute_reason	enum + free text	Tagged to a coarse category (amount_mismatch / not_my_loan / already_settled / other) for routing, full utterance kept for the human agent.

Entity	Type	Extraction notes
preferred_language	enum (en/hi)	Set on first detection of Hindi tokens; persists for the remainder of the call unless the caller switches again.
callback_window	datetime range	Validated against permitted calling hours before being accepted.

4. Tools / API Calls

All tools are exposed to the LLM as function-calling schemas but gated by state: the orchestrator only advertises the subset of tools legal in the current state, so the model cannot, for example, call `send_payment_link` before a PTP exists or `check_dnc_status` mid-negotiation for no reason.

Tool	Called from	Inputs	Outputs
<code>verify_customer</code>	S2	phone, dob, id_last4	verified (bool), attempts_left (int), account_id (string, only if verified)
<code>get_account_details</code>	S3	account_id	customer_name, loan_type, emi_amount, due_date, days_past_due
<code>check_dnc_status</code>	S1 (pre-check)	phone	is_dnc (bool), consent_channel (enum)
<code>log_promise_to_pay</code>	S4a	account_id, ptp_date, ptp_amount, channel	confirmation_id, status
<code>send_payment_link</code>	S6	account_id, channel (sms/whatsapp), amount	status, link_id, delivered (bool)
<code>log_dispute</code>	S4b	account_id, dispute_reason, notes	ticket_id
<code>escalate_to_agent</code>	S5	account_id, reason, transcript_summary, priority	ticket_id, queue
<code>mark_disposition</code>	S7	call_id, disposition_code, notes, ptp_ref (optional)	status (ack)

Sample schema (`send_payment_link`), illustrating the pattern used for all eight tools:

```
{
  "name": "send_payment_link",
  "parameters": {
    "account_id": "string, required",
    "channel": "enum[sms, whatsapp], required",
    "amount": "number, required, INR"
  },
  "returns": {
    "status": "enum[sent, failed]",
    "link_id": "string",
    "delivered": "boolean"
  }
}
```

5. Auth & Data Safety

5.1 Verifying identity before disclosure

- Two independent factors required before any account fact is spoken: date of birth and the last 4 digits of the loan account (or PAN), matched server-side by `verify_customer` — never compared by the LLM.

- Maximum 3 attempts, tracked by an orchestrator-side counter (not something the model can be argued into extending).
- The `auth_verified` flag is set only by a successful tool result and lives in orchestrator state, not in the prompt — it cannot be altered by anything the caller says.
- For this Vapi demo, debt facts are never placed in the pre-auth model context and the webhook blocks every account-data or payment tool until `auth_verified` is set by `verify_customer`. A production deployment additionally requires an output-policy filter before TTS to block any pre-auth debt disclosure attempt.

5.2 Protecting against third-party disclosure

- If verification fails or the person states they are not the customer, the bot never confirms whether the number belongs to a loan customer at all — it uses a neutral script (“I may have the wrong person — could you ask [name] to call us back?”) and logs `wrong_person`, ending the call without any financial content.
- No voicemail message includes the amount or the word “overdue” — only “this is Maya calling from Kapture Finance; please return our call using the official contact details”.
- For this demo, DOB plus ID-last-four is the mocked verification control. Production use requires the lender's compliance-approved identity-verification method (for example, an OTP or a stronger customer-authentication control); knowledge of these details alone must not be treated as third-party authorization.

5.3 Data handling

- PII (DOB, ID digits) is passed on a separate tool-call channel and masked in transcripts/logs; only the verified boolean and `account_id` persist in the conversation log.
- Call recordings and transcripts retained per the client's data-retention policy; access scoped to collections and QA roles only.

6. Guardrails & Compliance

6.1 Mandatory disclosures

- Opening line always states: bot identity (“this is Maya, an automated assistant from Kapture Finance”), the fact that the call may be recorded, and the neutral purpose (“calling regarding a confidential customer-service matter”) — before any question is asked.

6.2 Calling hours & consent

- Outbound dials only scheduled within RBI-permitted collection-call hours (08:00–19:00 local time); the orchestrator, not the LLM, enforces this at dial time.
- A `do_not_call` intent at any point immediately routes to S4e: the bot confirms the opt-out verbally, calls `mark_disposition` with a DNC code, and the number is suppressed from future dial lists — no further negotiation attempted on that call.

6.3 Tone & conduct

- System prompt hard-constrains tone: no threats, no repetition/pressure tactics, no implying legal action the client hasn't authorized, no contacting or disclosing to third parties, no raised urgency language beyond stating the actual due date.
- On hostile/abusive input, the bot de-escalates once (“I understand this is frustrating — I'm only here to help resolve your account”); a second instance of abuse ends the call politely rather than continuing to absorb it.

6.4 Hallucination & off-topic guardrails

- The model may only state figures returned by `get_account_details` — it is instructed never to compute, estimate, or infer an amount, and the response filter (§5.1) backstops this.
- Off-topic requests (new loans, unrelated complaints) get one polite redirect to the call’s purpose; a second attempt routes to `escalate_to_agent` rather than letting the model improvise an answer outside its remit.

7. Edge Cases

Case	Detection	Handling
Already paid	<code>already_paid</code> intent	Acknowledge, do not argue; call <code>escalate_to_agent</code> with <code>reason=payment_verification</code> so a human reconciles against the ledger; disposition <code>already_paid_claimed</code> .
Disputes the amount	<code>dispute</code> intent	Do not re-assert the figure repeatedly; <code>log_dispute</code> , offer to send a statement, escalate for review.
Requests do-not-call	<code>do_not_call</code> intent	Immediate opt-out flow (§6.2); no further negotiation.
Wrong number	<code>wrong_person</code> intent, pre- or post-auth	No financial disclosure (§5.2); mark <code>wrong_number</code> , remove from this customer’s dial attempt.
Voicemail / no input timeout	AMD at S0, or silence > 6s twice in a row mid-call	Voicemail: compliant callback message only. Mid-call silence: one gentle re-prompt, then a second timeout ends the call with disposition <code>no_response</code> .
Abusive caller	hostile intent, repeated	One de-escalation attempt, then graceful call end (§6.3).
Mid-call language switch (EN/HI)	language token detection each turn	STT/LLM/TTS all support both; <code>preferred_language</code> updates immediately and the bot continues in the new language without restarting the flow.
Tool/API failure or timeout	webhook error or > 600 ms with no response	Bot plays a natural filler, retries once; on repeated failure it apologizes, logs a <code>technical_failure</code> disposition, and offers a human callback rather than guessing at account data.

8. Escalation & Disposition

8.1 When the bot hands off to a human

- Disputes of amount/ownership, hardship cases beyond a scripted standard offer, an already-paid claim, repeated hostility after one de-escalation attempt, low intent-classification confidence for two consecutive turns, and any explicit “talk to a person” request.

8.2 Every call ends with a logged disposition

S7 is the single mandatory exit for every call path (see Figure 2) — there is no route that reaches S8 without first calling `mark_disposition`. Disposition codes:

Code	Meaning
<code>ptp_captured</code>	Promise-to-pay logged, payment link sent
<code>already_paid_claimed</code>	Customer claims payment made — escalated for reconciliation
<code>dispute_logged</code>	Amount/ownership disputed — escalated

Code	Meaning
hardship_escalated	Cannot-pay / hardship — escalated for restructuring options
dnc_opt_out	Customer opted out of future calls
wrong_number	Not the customer — no disclosure made
no_disclosure_unverified	Auth failed 3x — call ended without disclosure
no_response	Silence/no input after re-prompt
voicemail_left	Compliant voicemail message left
technical_failure	Tool/API failure prevented completion

9. Observability

9.1 What is logged per call

- Full transcript (PII-masked per §5.3), state-transition trace with timestamps, every tool call + its latency and result, final disposition code, and per-hop latency (STT/LLM/TTS) for the whole call.

9.2 Metrics tracked

Metric	Why it matters
Containment rate	% of calls resolved without human escalation — the core efficiency metric for the bot.
PTP rate	% of authenticated calls ending in a logged promise-to-pay — the core business outcome.
PTP kept rate (downstream)	% of PTPs actually paid by the promised date — validates whether the negotiation is producing real commitments, not just agreeable ones.
Auth success / failure rate	Tracks whether verification questions are well-calibrated — too many failures may mean genuine customers are getting locked out.
Avg. turn latency (p50/p95)	Directly affects perceived call quality and containment.
Drop / hang-up rate by state	Flags exactly where in the flow customers abandon the call — the most actionable debugging signal.
Escalation rate by reason	Shows which intents the bot cannot yet resolve, prioritizing prompt/flow improvements.
DNC compliance latency	Time between an opt-out request and suppression in the dialer — a compliance-critical SLA.

10. Assumptions & Out of Scope

- Payment gateway, SMS/WhatsApp, and CRM/loan-ledger integrations are mocked webhooks for this exercise, matching real response shapes but not connected to live systems.
- DOB plus last-4-ID is a mock-only exercise control. Production must use a lender-approved authentication method; this build does not authorize third parties based on knowledge of these details.
- Calling-hours enforcement and DNC-list suppression are described as orchestrator-level controls; the actual dialer/scheduling system sits outside this bot and is assumed to exist at Kapture Finance.
- Language support is scoped to English and Hindi per the brief; other regional languages are out of scope.
- Legal/regulatory language (e.g. exact RBI Fair Practices Code wording) should be reviewed by Kapture's compliance team before production use — this HLD implements the spirit of those norms, not a legal sign-off.